Block practice · Sorting and searching: bubble sort, merge sort, binary search

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Bubble sort, one pass at a time

Merge sort: split, then merge

Binary search step by step

Compare sorts and searches

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Practice workbook · Bubble sort, one pass at a time

Showing the state of a list after each bubble sort pass, and counting swaps or passes.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-08

1. Question 3 · part a.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

1. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b.i · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b.i only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2. Question 5 · part b.i · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b.i only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

.....

.....

.....

.....

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P 7 2 4 0 6 A 0 1 1 1 6

2. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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3. Question 3 · part a.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

4. Question 3 · part a.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a.iii only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

.....

Justification

.....

.....

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4. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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Practice workbook · Merge sort: split, then merge

Showing the split and merge stages of a merge sort on a given list, ascending or descending.

2 question parts: 1 worked-example retry and 1 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-09

1. Question 2 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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1. Question 2 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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DO NOT WRITE IN THIS AREA

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1. Question 2 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

(c) A merge sort algorithm can be used to sort a list into ascending order.

The list 9, 1, 7, 6, 3, 5, 2, 8 needs to be sorted.

Complete the merge sort using the space provided.

(5)

9, 1, 7, 6, 3, 5, 2, 8

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1, 2, 3, 5, 6, 7, 8, 9

(Total for Question 2 = 19 marks)

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1. Question 2 · part c · 5 marks · Working and answer

Working and answer

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2. Question 5 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

5 A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

Figure 5 shows the lists created at the end of the splitting process.

Each list is a single tuna species.

Bigeye	Blackfin	Albacore	Longtail	Bluefin

Figure 5

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

(2)

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DO NOT WRITE IN THIS AREA

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2. Question 5 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

Figure 6 shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

Figure 6

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

Line number with error	
Corrected line of pseudocode	

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12

P 7 2 9 3 8 A 0 1 2 1 6

2. Question 5 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Binary search step by step

Stepping through binary search on a sorted list (order of items examined, low/high/mid values), worst-case number of checks, and why it fails on unsorted data.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-10

1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

5 Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

Figure 6

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

10	12	13	14	15	16	17	18	19	20

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

high index	low index	midpoint

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1. Question 5 · part b · 4 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

(c) **Figure 7** shows the **Sales.txt** file. It stores sales information.

```
264,140,120,284,192
420,377,435,376,392
619,589,606,586,600
799,811,788,814,788
982,1007,1013,989,1009
```

Figure 7

A program is required to calculate and display:

- a subtotal for each line of sales
- a grand total for all the sales in the file.

Figure 8 shows the intended output from the program.

```
1000
2000
3000
4000
5000
Grand total: 15000
```

Figure 8

Open **Q05c** in the code editor.

Amend the code to produce the intended output.

You must use the structure and variables given in **Q05c** to complete the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 14 marks)

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1. Question 5 · part b · 4 marks · Working and answer

Working and answer

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2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12

P 6 6 4 9 0 A 0 1 2 1 6

2. Question 4 · part b.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

(iv) Give the name of an algorithm that could be used to sort a list.

(1)

(Total for Question 4 = 15 marks)

Working and answer

OLevels.com SCHOOL

3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 4 9 0 A 0 1 2 1 6

3. Question 4 · part b.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

.....

(iv) Give the name of an algorithm that could be used to sort a list.

(1)

.....

(Total for Question 4 = 15 marks)

Working and answer

OLevels.com SCHOOL

4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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OLevels.COM SCHOOL

4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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4. Question 2 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(c) A merge sort algorithm can be used to sort a list into ascending order.

The list 9, 1, 7, 6, 3, 5, 2, 8 needs to be sorted.

Complete the merge sort using the space provided.

(5)

9, 1, 7, 6, 3, 5, 2, 8

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1, 2, 3, 5, 6, 7, 8, 9

(Total for Question 2 = 19 marks)

4. Question 2 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Compare sorts and searches

How linear search, binary search, bubble sort and merge sort work in words, and their relative advantages, drawbacks, memory use and suitability for a given data set.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com — SCHOOL —

Type ID: CSC-T1-11

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

.....

Justification

.....

.....

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) A program is required to calculate and display the average age of members taking part in an activity.

Figure 3 shows a flowchart for the algorithm.

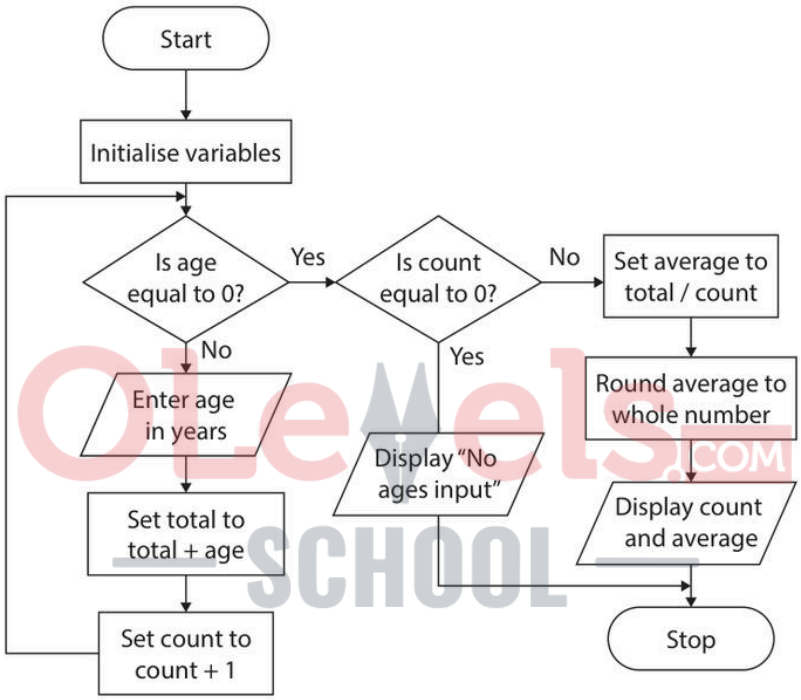


Figure 3

Open **Q03c** in the code editor.

Write a program to implement the logic of the flowchart.

Use the data given in **Figure 4** to test your program.

Test data	Expected output
7 8 9 10 11 0	You entered 5 ages; the average age is 9.
16 16 14 0	You entered 3 ages; the average age is 15.
0	No ages input

Figure 4

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 3 = 11 marks)

1. Question 3 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

3 Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(6)

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

(2)

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 14 marks)

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2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.i only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

She is creating a stock control program.

(a) Each item of stock will have a ten-character code.

Each code consists of three parts.

Part	Description
Characters 1 to 3	Uppercase letters
Characters 4 to 9	Numbers between 1 and 9 inclusive
Character 10	'e' or 'o' indicating whether the sum of the numbers is even or odd

WAS123456o, FDG232324e and VAC129222e are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test, give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is the last character 'o' or 'e'?	ABC654321v

(b) Cerys will need an algorithm to search the table of stock data.

(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

P 6 1 8 8 5 A 0 9 1 6

3. Question 5 · part b.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.i only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

(c) Cerys will use functions and procedures in her program.

(i) One similarity between a function and a procedure is that they are both subprograms.

State **two** other similarities.

(2)

1 .....

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2 .....

.....

(ii) Explain **one** difference between a function and a procedure.

(2)

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(Total for Question 5 = 12 marks)

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3. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.ii only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

She is creating a stock control program.

(a) Each item of stock will have a ten-character code.

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Part	Description
Characters 1 to 3	Uppercase letters
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(4)

Validation test	Erroneous data
Is the last character 'o' or 'e'?	ABC654321v

(b) Cerys will need an algorithm to search the table of stock data.

(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

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4. Question 5 · part b.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.ii only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

(c) Cerys will use functions and procedures in her program.

(i) One similarity between a function and a procedure is that they are both subprograms.

State **two** other similarities.

(2)

1 .....

.....

2 .....

.....

(ii) Explain **one** difference between a function and a procedure.

(2)

.....

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(Total for Question 5 = 12 marks)

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4. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b.i · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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5. Question 4 · part b.i · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

5. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iv only.

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  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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P 6 6 4 9 0 A 0 1 1 1 6

6. Question 4 · part b.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iv only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
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13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
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3	bar13	
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5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

6. Question 4 · part b.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iv only.

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(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

.....

(iv) Give the name of an algorithm that could be used to sort a list.

(1)

.....

(Total for Question 4 = 15 marks)

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Working and answer

7. Question 5 · part b.ii · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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7. Question 5 · part b.ii · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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P 7 2 4 0 6 A 0 1 1 1 6

7. Question 5 · part b.ii · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

(c) Describe the steps a linear search algorithm takes to find a search item.

(3)

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(Total for Question 5 = 17 marks)

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Working and answer

8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

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(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

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(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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P 7 2 4 0 6 A 0 1 1 1 6

8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

(c) Describe the steps a linear search algorithm takes to find a search item.

(3)

DO NOT WRITE IN THIS AREA

(Total for Question 5 = 17 marks)

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Working and answer